Oxford Cambridge and RSA – style practice paper

GCSE (9–1) Computer Science

J277/01 Computer systems

Topic 1.2: Memory and storage

Candidate forename		**Candidate surname**	
Centre number		**Candidate number**	

Practice questions in the style of OCR GCSE Computer Science (J277). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 120 minutes (roughly 1 minute per mark)
Instructions	• Use black ink. HB pencil may be used for graphs and diagrams only. • Answer **all** the questions. • Write your answer to each question in the space provided. • Where the question asks you to write code or pseudocode, you may use any suitable high-level language or the OCR Reference Language; marks are for the logic, not exact syntax. • Do **not** use a calculator or any other electronic device unless told to.
Information	• The total mark for this paper is **108**. • The marks for each question are shown in brackets [ ]. • Quality of extended responses will be assessed in questions marked with an asterisk (*). • Where a question asks you to *state* or *give*, a short answer is enough; *describe* and *explain* need more detail.

Name: _____ Date: _____ Mark: _____ / 108

1 A laptop has both RAM and ROM.

(a) State **two** differences between RAM and ROM. **[2]**

(b) State what is stored in ROM in a typical computer. **[1]**

(c) The laptop has 4 GB of RAM. Several programs are open and RAM is full. Describe how the operating system uses **virtual memory** to allow the user to open another program. **[3]**

(d) Explain why using virtual memory can make the computer run slowly. **[2]**

(e) Give **one** way the user could reduce the use of virtual memory. **[1]**

2 A company is choosing secondary storage devices. The table shows information about four types of device.

(a) Complete the table by writing the type of storage (**magnetic**, **optical** or **solid state**) used by each device. **[3]**

Device	Type of storage	Typical capacity	Moving parts?
Hard disk drive (HDD)		1 TB	Yes
DVD		4.7 GB	Yes
Solid state drive (SSD)		500 GB	No
USB flash drive		64 GB	No

(b) A photographer needs to store 800 GB of photographs permanently at the lowest cost per GB. Recommend a suitable device from the table and justify your choice. **[3]**

(c) A teacher wants a device to carry a presentation between school and home in a pocket. Explain why a USB flash drive is more suitable than a hard disk drive. **[3]**

(d) State **two** advantages of SSDs over hard disk drives. **[2]**

(e) Explain how data is stored on an optical disc. **[3]**

3 Computers store all data in binary.

(a) State how many bits there are in a byte. **[1]**

(b) State what is meant by a **nibble**. **[1]**

(c) Put these units in order of size, starting with the smallest: **gigabyte, kilobyte, terabyte, megabyte**. **[2]**

(d) A file is 4500 KB. Convert this to megabytes. Use 1 MB = 1000 KB. **[1]**

Answer _____ MB

(e) A memory card has a capacity of 32 GB. Each photo is 8 MB. Calculate how many photos fit on the memory card. Use 1 GB = 1000 MB. **[2]**

Answer _____ photos

(f) Explain why computers use binary to represent data. **[2]**

4 Number conversions.

(a) Convert the binary number **10110101** to denary. **[2]**

Answer _____

(b) Convert the denary number **203** to 8-bit binary. **[2]**

Answer _____

(c) Convert the binary number **10110101** to hexadecimal. **[2]**

Answer _____

(d) Convert the hexadecimal number **3F** to 8-bit binary. **[2]**

Answer _____

(e) Convert the hexadecimal number **A7** to denary. **[2]**

Answer _____

(f) Give **one** reason why programmers use hexadecimal instead of binary. **[1]**

_____

_____

5 Binary arithmetic and shifts.

(a) Add the two 8-bit binary numbers **01101101** and **00110111**. Show your working. **[3]**

Answer _____

(b) An 8-bit register adds **11001000** and **01100100**. Explain what happens. **[2]**

_____

_____

_____

(c) Perform a binary shift **two places to the left** on **00001101**. **[2]**

Answer _____

(d) State the effect on the denary value of shifting a binary number two places to the left. **[1]**

_____

(e) Perform a binary shift **three places to the right** on **10110000** and give the denary value of the result. **[3]**

Answer _____

6 Text is stored using character sets.

(a) State what is meant by a **character set**. **[1]**

(b) In ASCII the capital letter **A** has the denary code 65. State the denary ASCII code for **H**. **[1]**

(c) The ASCII code for **a** is 97. Write the 8-bit binary for the code of the lower case letter **e**. **[2]**

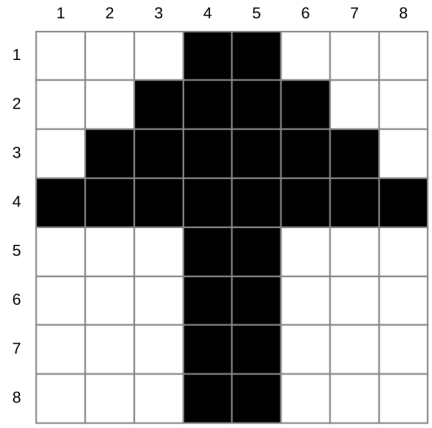
Answer _____

(d) Explain why Unicode is used instead of ASCII for text on the internet. **[3]**

(e) A text file contains 2000 characters and uses 8-bit ASCII. Calculate the size of the file in kilobytes (1 KB = 1000 bytes). **[2]**

Answer _____ KB

7 The diagram shows an 8 × 8 black and white image. Each pixel is stored as 1 bit: black = 1 and white = 0.



(a) Write the binary for **row 3** of the image. **[1]**

(b) State the size of the image file in bytes, ignoring metadata. **[2]**

Answer _____ bytes

(c) The image is changed to use 16 different colours. Calculate the number of bits needed per pixel. **[2]**

Answer _____ bits

(d) State the term for the number of bits used for the colour of each pixel. **[1]**

(e) Explain the effect on the image of **increasing the resolution**. **[2]**

(f) Give **one** example of **metadata** stored with an image file. **[1]**

8 A photograph is 1920 pixels wide and 1080 pixels high. It uses a colour depth of 24 bits per pixel.

(a) Calculate the file size of the image in bytes. Show your working. Ignore metadata. **[3]**

Answer _____ bytes

(b) State the file size in megabytes (1 MB = 1 000 000 bytes). Give your answer to 2 decimal places. **[1]**

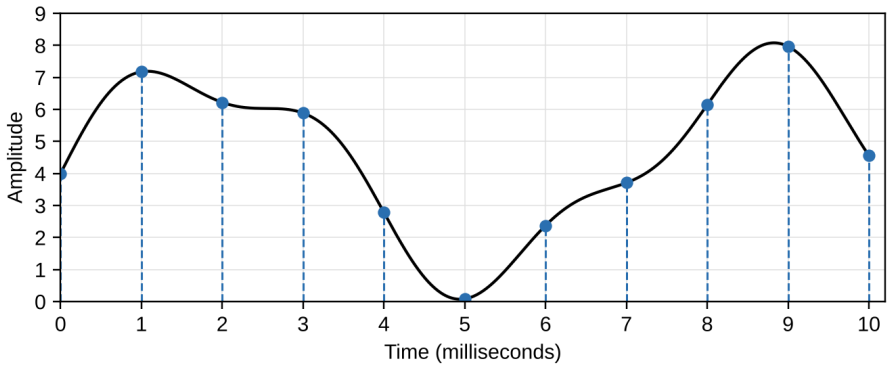
Answer _____ MB

(c) The photograph is compressed using lossy compression. Describe how **lossy** compression reduces the file size. **[2]**

(d) Explain why lossless compression is more suitable than lossy compression for a computer program file. **[2]**

(e) Give **one** advantage of compressing files before sending them by email. **[1]**

9 The graph shows a sound wave that has been sampled. The vertical dashed lines show each sample point.



(a) What is meant by **sample rate**? **[1]**

(b) The sound was sampled every 1 millisecond. Calculate the sample rate in hertz. (1 second = 1000 milliseconds) **[2]**

Answer _____ Hz

(c) Write down the first **four** sample values, to the nearest whole number, starting at 0 ms. **[2]**

(d) Explain how increasing the sample rate affects the sound file. **[2]**

(e) A song lasts 3 minutes (180 seconds). It is sampled at 44 100 Hz with a bit depth of 16 bits, in mono. Calculate the file size in megabytes (1 MB = 1 000 000 bytes). Show your working. **[4]**

Answer _____ MB

10 Extended response questions.

(a) *A company stores customer records, high-definition video adverts and a large number of backup files. Discuss the secondary storage the company could use. Your answer should compare magnetic, optical and solid state storage. **[8]**

(b) *A music streaming service stores songs as compressed sound files. Discuss the effect of sample rate, bit depth and compression on the quality and file size of the sound files. **[8]**

END OF QUESTIONS

MARK SCHEME – Topic 1.2: Memory and storage

How to use this mark scheme: mark your own work in a different colour. A bullet (•) in the Answer column is a separate marking point worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed. **Accept** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. For programming questions, any valid solution that works is acceptable and syntax errors are not penalised if the intent is clear. The code in the answers uses the OCR Reference Language style.

Question	Answer	Marks	Guidance
1(a)	• RAM is volatile / contents lost when power is off, ROM is non-volatile / retains contents ; • RAM can be read from and written to, ROM is (normally) read only ; • RAM stores programs / data currently in use, ROM stores start-up instructions (boot program) ;	2 AO1 1.2.1	Any two differences. Must be a comparison.
1(b)	the bootstrap / start-up program (BIOS) ;	1 AO1 1.2.1	Accept: instructions needed to start the computer.
1(c)	• part of the secondary storage (hard drive / SSD) is used as extra memory ; • data / pages not currently needed are moved from RAM to virtual memory ; • space in RAM is freed so the new program can be loaded ; (data moved back when needed)	3 AO2 1.2.1	
1(d)	• secondary storage is slower to access than RAM ; • data must be swapped (paged) between RAM and virtual memory repeatedly ;	2 AO2 1.2.1	
1(e)	install / add more RAM / close programs not being used ;	1 AO3 1.2.1	
2(a)	• hard disk drive: magnetic ; • DVD: optical ; • SSD / USB flash drive: solid state ;	3 AO1 1.2.2	Award 1 mark for each of three correct types (SSD and USB flash count together as one mark).
2(b)	• hard disk drive (magnetic) ; • large capacity (1 TB or more) ; • cheap per GB / low cost ;	3 AO3 1.2.2	Accept an SSD only if justified by speed / durability and capacity.
2(c)	• flash drive is smaller / lighter / more portable ; • no moving parts so more durable / less likely to be damaged by knocks ; • suitable capacity for a presentation / plugs directly into a USB port ;	3 AO2 1.2.2	
2(d)	• faster read / write speeds ; • no moving parts so more durable / reliable ; • silent / lower power consumption / less heat ;	2 AO1 1.2.2	Any two.
2(e)	• data is stored as pits and lands ; • on a spiral track on the disc surface ; • a laser is shone at the surface and the reflection read: a pit scatters the light, a land reflects it (representing 0 and 1) ;	3 AO1 1.2.2	
3(a)	8	1 AO1 1.2.3	
3(b)	4 bits ;	1 AO1 1.2.3	
3(c)	• kilobyte, megabyte, gigabyte, terabyte (correct order) ; • all four in the correct order = 2 marks ;	2 AO1 1.2.3	2 marks for all correct; 1 mark for exactly one pair swapped.

Question	Answer	Marks	Guidance
3(d)	4.5 MB	1 AO2 1.2.3	
3(e)	• 32 × 1000 = 32 000 MB ; • 32 000 ÷ 8 = 4000 photos ;	2 AO2 1.2.3	
3(f)	• computers are made of switches / transistors / logic gates ; • which have two states (on / off, 1 / 0) ;	2 AO1 1.2.3	
4(a)	• 128 + 32 + 16 + 4 + 1 ; • 181 ;	2 AO2 1.2.4	Award 1 mark for correct place values / working.
4(b)	• 128 + 64 + 8 + 2 + 1 ; • 11001011 ;	2 AO2 1.2.4	
4(c)	• 1011 = B and 0101 = 5 ; • B5 ;	2 AO2 1.2.4	
4(d)	• 3 = 0011 and F = 1111 ; • 00111111 ;	2 AO2 1.2.4	
4(e)	• A = 10 so (10 × 16) + 7 ; • 167 ;	2 AO2 1.2.4	
4(f)	easier to read / write / remember / fewer digits / less chance of error ;	1 AO1 1.2.4	Accept: it is a shorthand for binary.
5(a)	• correct column addition with carries ; • 10100100 ; • (allow denary check 109 + 55 = 164) ;	3 AO2 1.2.4	Award 2 marks for the correct answer; 1 mark for the answer with a single error in a column.
5(b)	• the answer is 300 which needs 9 bits ; • overflow error: the result is too big for 8 bits (an extra bit is lost) ;	2 AO3 1.2.4	
5(c)	• 00110100 ; • zeros added on the right ;	2 AO2 1.2.4	
5(d)	the number is multiplied by 4 ;	1 AO1 1.2.4	
5(e)	• 00010110 ; • 22 ; • (or divided by 8: 176 ÷ 8) ;	3 AO2 1.2.4	
6(a)	a (agreed) set of characters with a binary code for each ;	1 AO1 1.2.5	
6(b)	72	1 AO2 1.2.5	A = 65, so H is 7 later.
6(c)	• e = 101 ; • 01100101 ;	2 AO2 1.2.5	
6(d)	• Unicode uses more bits per character (16 / 32 bits) ; • so can represent many more characters / symbols / emojis / other languages ; • ASCII (7 or 8 bits) has only 128 / 256 characters ;	3 AO1 1.2.5	
6(e)	• 2000 × 8 = 16 000 bits / 2000 bytes ; • 2 KB ;	2 AO2 1.2.5	
7(a)	01111110	1 AO2 1.2.6	
7(b)	• 8 × 8 = 64 bits ; • 8 bytes ;	2 AO2 1.2.6	

Question	Answer	Marks	Guidance
7(c)	• 16 = 2^4 ; • 4 bits ;	2 AO2 1.2.6	
7(d)	colour depth	1 AO1 1.2.6	
7(e)	• more pixels (so more detail / better quality) ; • larger file size ;	2 AO2 1.2.6	
7(f)	any from: image dimensions / width and height ; colour depth ; date taken ; camera type ; location / GPS ; file format ;	1 AO1 1.2.6	
8(a)	• 1920 × 1080 = 2 073 600 pixels ; • × 24 = 49 766 400 bits ; • ÷ 8 = 6 220 800 bytes ;	3 AO2 1.2.6	
8(b)	6.22 MB	1 AO2 1.2.6	
8(c)	• permanently removes some data / detail ; • that is not noticeable / less important to the human eye ;	2 AO1 1.2.7	
8(d)	• lossless allows the original file to be rebuilt exactly ; • lossy would remove data so the program would not work / be corrupted ;	2 AO2 1.2.7	
8(e)	smaller file so quicker to send / uses less bandwidth / fits attachment limit / less storage ;	1 AO1 1.2.7	
9(a)	the number of samples taken per second ;	1 AO1 1.2.6	
9(b)	• 1000 ÷ 1 ; • 1000 Hz ;	2 AO2 1.2.6	
9(c)	• 4, 7, 6, 6 ; • (all four correct = 2 marks; two or three correct = 1 mark) ;	2 AO3 1.2.6	Allow ±0.5 tolerance on each value.
9(d)	• more accurate / closer to the original sound / better quality ; • larger file size ;	2 AO2 1.2.6	
9(e)	• 44 100 × 16 × 180 ; • = 127 008 000 bits ; • ÷ 8 = 15 876 000 bytes ; • 15.88 MB (15.876) ;	4 AO2 1.2.6	

Question	Answer	Marks	Guidance
10(a)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • magnetic (HDD): very large capacity; low cost per GB; suitable for backups and archives; slower, moving parts so less durable, uses more power • solid state (SSD / flash): fast access / no moving parts / durable / low power; more expensive per GB; suitable for the working files and customer records needing quick access • optical (DVD / Blu-ray): low capacity, cheap discs, portable and can be posted; slow; easily scratched; suitable for delivering finished video adverts to clients or one-off archive copies • a combination is sensible: SSD for active databases, HDD arrays for backups, optical not suitable for large volumes • consideration of capacity, speed, portability, durability, reliability and cost in context • balanced conclusion	8 AO1 / AO2 / AO3 1.2.2	
10(b)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • sample rate: samples per second; higher = more accurate wave, better quality, larger file • bit depth: number of bits per sample; higher = more precise amplitude values, better quality, larger file • file size = sample rate × bit depth × duration (× channels) • lossy compression removes data the ear is less likely to notice; much smaller file; permanent loss of quality; suitable for streaming / limited bandwidth • lossless compression keeps all data; file is smaller but not as small as lossy; the original can be restored; used for archives / audiophiles • trade-off between quality and storage / bandwidth costs; balanced conclusion about the best settings for streaming	8 AO1 / AO2 / AO3 1.2.6 / 1.2.7	

Where marks are lost in Topic 1.2

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Show your working** for every conversion and file-size calculation. The method marks are easy to bank even if the arithmetic slips.
- ☐ **Binary place values:** 128 64 32 16 8 4 2 1. Write them above the number before converting.
- ☐ **Hex:** convert each 4-bit nibble separately (1011 = B, 0101 = 5). Don't convert the whole byte to denary first.
- ☐ **File sizes:** image = width × height × colour depth; sound = sample rate × bit depth × seconds. Divide by 8 for bytes. Say which unit you've used.
- ☐ **Overflow:** if an 8-bit addition produces a 9th bit, the result is too big to store. Use the word 'overflow'.
- ☐ **Shifts:** left shift ×2 per place, right shift ÷2 per place. Always keep the number of bits the same.
- ☐ **Lossy vs lossless:** lossy = data permanently removed, smaller; lossless = original restored exactly. Link choice to the file type.
- ☐ **Storage questions:** compare using capacity, speed, portability, durability, reliability and cost — and link each to the scenario.